

Survivor Detection & Rescue Coordination Guide (RAG-Optimized)

- 1. Overview

This document defines operational rules for autonomous drone swarms performing survivor detection, validation, and rescue coordination in disaster environments (earthquakes, collapses, urban failures).

Objectives:

- Detect human survivors using multi-sensor fusion
- Validate detections with confidence scoring
- Prevent duplicate detections across swarm units
- Coordinate multi-drone verification
- Prioritize rescue actions dynamically
- Operate fully autonomously under MCP/swarm control

This guide is intended for machine agents, not human operators.

- 2. Survivor Detection Pipeline

Each drone executes a standardized pipeline:

Step 1: Sector Assignment & Navigation

- Receive sector from Command Agent
 - Navigate via `move_to(x, y)`
 - Initialize environmental scan mode
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Step 2: Multi-Sensor Activation

Drones activate all available sensors:

- Thermal imaging (`thermal_scan`)
- Visual capture (`capture_image`)
- Motion detection (if available)
- Structural gap / debris analysis

Outputs:

- Thermal heatmap
 - Visual frame set
 - Motion anomaly signals
 - Environmental interference map
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Step 3: Candidate Detection

A survivor candidate is flagged when anomalies are detected.

Detection Signals:

- Human-shaped thermal signature
- Localized heat anomaly in debris
- Motion in collapsed structures
- Repeating heat patterns (e.g., breathing)

Candidate Metadata:

- Coordinates (x, y)
 - Detection type: thermal / motion / hybrid
 - Initial confidence score (0.0–1.0)
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- 3. Validation & Confidence Scoring

All detections require validation to reduce false positives.

3.1 Validation Criteria

A detection is valid if ANY apply:

- Thermal range matches human body temperature
 - Movement detected in debris field
 - Heat signature persists across ≥ 2 scans
 - Multi-sensor confirmation (thermal + visual)
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3.2 Confidence Scoring Model

- High Confidence (0.80 – 1.00)

Strong confirmation indicators:

- Clear human thermal silhouette
- Stable heat signature over time
- Movement or breathing pattern detected

Actions:

- Immediate escalation to Command Agent
 - Assign nearby drones for reinforcement
 - Continuous tracking enabled
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● **Medium Confidence (0.50 – 0.79)**

Partial confirmation:

- Occluded or partially blocked thermal signal
- Weak/intermittent detection
- No confirmed motion

Actions:

- Dispatch secondary drone for verification
 - Re-scan within 10–15 minutes
 - Require multi-angle confirmation
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● **Low Confidence (0.20 – 0.49)**

Uncertain signals:

- Ambiguous heat anomalies
- Possible environmental interference
- Single-sensor detection only

Actions:

- Mark as “unconfirmed candidate”
 - Schedule periodic re-scan
 - Do not allocate rescue priority
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● **Invalid (< 0.20)**

False positives:

- Environmental heat noise
- Fire, machinery, or debris artifacts
- No spatial or temporal consistency

Actions:

- Discard from active tracking system
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3.3 Cooperative Verification Protocol

To ensure accuracy:

- Nearest drone performs first confirmation
- Second drone verifies from alternate angle
- Third drone provides overwatch (if available)

This forms a triangulated verification system to reduce false positives.

• 4. Rescue Prioritization System

Survivors are ranked based on urgency, confidence, and environmental risk.

● Priority 1 — Immediate Rescue

Conditions:

- Multiple survivors detected OR
- Confidence > 0.80 AND
- High structural instability detected

Actions:

- Lock sector (no reassignment)
 - Continuous drone surveillance
 - Immediate relay to rescue teams
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● Priority 2 — High Urgency

Conditions:

- Single confirmed survivor
- Confidence 0.50–0.79
- Environment stable

Actions:

- Assign nearest drone for monitoring
 - Map potential extraction routes
 - Maintain periodic updates
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● Priority 3 — Monitoring

Conditions:

- Weak or unconfirmed signals
- No validation yet

Actions:

- Scheduled re-scanning only
 - No resource locking
 - Passive tracking mode
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- 5. Swarm Coordination Rules

All drones must coordinate via shared state map.

Core Rules:

- Always verify before escalating detections
 - Prevent duplicate survivor tagging
 - Share confirmed detections in real time
 - Avoid redundant scanning of same coordinates
 - Optimize battery for high-confidence zones
 - Dynamically reassign drones based on priority shifts
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- 6. Key Operational Principles

The swarm operates under strict autonomy constraints:

Mandatory Principles:

- Validation-first decision making
- Confidence-based prioritization
- Multi-drone verification for critical detections
- Efficient energy usage (battery-aware routing)
- Shared global state awareness
- Continuous adaptive mission planning